ALI EL HABCHI

Assistant Professor | Computer Vision & Applied AI

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AI researcher, engineer, and educator with a PhD in Computer Science and AI. Experience spans UAV perception, embedded inference, multilingual NLP, and production ML workflows. Teaching connects computer vision and software engineering with practical implementation.

PROFESSIONAL EXPERIENCE

Assistant Professor in AI & Software Engineering | ENIAD Berkane

October 2025 - Present | Mohammed Premier University, Morocco

- Teach image processing and computer vision, DevOps/MLOps, software modelling and UML, and related software engineering subjects.
- Develop practical coursework and demonstrations, supervise student projects, and participate in project evaluation and academic juries.

Senior Machine Learning Engineer | Onclusive

May 2024 - September 2025 | Rabat, Morocco

- Worked on multilingual NLP tasks including sentiment, named-entity recognition, topic detection, keyword extraction, clustering, and summarization.
- Contributed to transformer model migration and inference workflows, including tokenization, chunking, evaluation, quantization, and AWS-based pipelines.
- Worked with Python, Hugging Face and PyTorch, connecting model development with deployment and operational requirements.

Machine Learning Engineering | Onclusive Social (formerly Digimind)

October 2023 - May 2024 | Morocco

- Developed visual brand-analysis models using YOLOv5 for logo detection and ResNet50V2 for classification.
- Worked on dataset management and reproducible delivery using DVC, MLflow, GitLab CI, Docker, Kubernetes, and AWS.

Senior R&D / Computer Vision Engineer | ATLAN Space

August 2018 - September 2023 | Rabat, Morocco

- Trained and deployed object detection and classification systems for aerial applications, including obstacles, vehicles, people, and boats.
- Implemented inference on Jetson Nano and Xavier AGX with OpenCV DNN, TensorRT and ONNX runtimes; developed C++ inference components.
- Developed optical-flow and multi-view approaches for obstacle perception and depth estimation, and target-geolocation software using OpenGL and terrain geometry.
- Contributed to drone software, communication and sensing components, photogrammetry workflows, and Azure-based development environments.

RESEARCH, EDUCATION & PRACTICE

EDUCATION

PhD in Computer Science & Artificial Intelligence

2025 | ENSA Oujda, Mohammed Premier University

- Doctoral research on UAV navigation through computer vision and deep learning, including obstacle perception, depth and motion cues, and aerial geolocation.

Master of Science in Software Engineering | 2018
Faculty of Sciences, Mohammed Premier University, Oujda

Licence in Mathematics and Informatics | 2016
Faculty of Sciences, Mohammed Premier University, Oujda

SELECTED PUBLICATIONS

2023 Evolutionary algorithm for optimized CNN architecture search applied to real-time boat detection in aerial images
Journal of Real-Time Image Processing 20, article 78. Co-author.

2022 Social distance monitoring using YoloV4 on aerial drone images
E3S Web of Conferences 351, 01035. First author.

2020 CGA: A New Approach to Estimate the Geolocation of a Ground Target from Drone Aerial Imagery
International Conference on Intelligent Computing in Data Sciences (ICDS). First author.

2020 CNN Adaptations for Boat Detection in Aerial Images Tested on Yolo v2
International Symposium on Advanced Electrical and Communication Technologies (ISAECT). Co-author.

TECHNICAL PRACTICE

Learning & perception: Python, PyTorch, TensorFlow/Keras, OpenCV, NumPy, Hugging Face Transformers, object detection and classification, camera geometry, visual motion and depth.

Inference & delivery: TensorRT, ONNX, NVIDIA Jetson, quantization, AWS, Docker, Kubernetes, DVC, MLflow, GitLab CI, FastAPI.

Software & geometry: C++, Java, SQL, Git, Linux, OpenGL, software modelling and UML.

SELECTED TRAINING & LANGUAGES

DevOps for Data Scientists (2023); Advanced AI: Transformers for Computer Vision (2023); Introduction to Generative AI (2023).

Arabic: native | **French:** professional proficiency | **English:** professional proficiency